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Probability Distributions**Ex 11.1****Question 1.**

Suppose X is the number of tails occurred when three fair coins are tossed once simultaneously.

Find the values of the random variable X and number of points in its inverse images.

Solution:

Let X is the random variable denotes the number of tails when three coins are tossed simultaneously.

Sample space $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

Therefore ' X ' takes the values 0,1,2,3

i.e., $X(HHH)=0; X(HHT, HTH, THH)=1; X(HTT, THT, TTH)=2; X(TTT)=3$

Question 2.

In a pack of 52 playing cards, two cards are drawn at random simultaneously. If the number of black cards drawn is a random variable, find the values of the random variable and number of points in its inverse images.

Solution:

Total number of playing cards = 52

Number of Black cards = 26

Number of Non-black (or) Red cards = 26

Let ' X ' be the random variable denotes the number of black cards. Since two black cards are drawn, ' X ' takes the values 0,1,2

$X(\text{Non-black Cards}) = X(26C1 \cdot 25C1) = X(650) = 0$

$X(1 \text{ Black Card}) = X(26C1 \cdot 26C0) = X(26) = 1$

$X(2 \text{ Black Cards}) = X(26C1 \cdot 25C1) = X(650) = 2$

Question 3.

An urn contains 5 mangoes and 4 apples. Three fruits are taken at random. If the number of apples taken is a random variable, then find the values of the random variable and number of points in its inverse images.

Solution:

Number of mangoes = 5

Number of Apples = 4

Total number of fruits = 9

Let ' X ' be the random variable denotes the number of apples taken, then it takes the values 0,1,2,3

$X(MMM) = 0$

$X(AMM \text{ (or) } MAM \text{ (or) } MMA) = 1$

$X(AAM \text{ (or) } AMA \text{ (or) } MAA) = 2$

$X(AAA) = 3$

Question 4.

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Two balls are chosen randomly from an urn containing 6 red and 8 black balls. Suppose that we win 15 for each red ball selected and we lose 10 for each black ball selected.

X denotes the

winning amount, then find the values of X and number of points in its inverse images.

Solution:

Number of red balls = 6

Number of black balls = 8

' X ' is the random variable denotes the winning amount.

Therefore The values of ' X ' are 0,15,30

i.e., $X(BB) = 0$

$$X(RB)=15+0=15$$

$$X(RR)=15+15=30$$

Question 5.

A six sided die is marked ' 2 ' on one face, ' 3 ' on two of its faces, and ' 4 ' on remaining three faces. The die is thrown twice. If X denotes the total score in two throws, find the values of the random variable and number of points in its inverse images.

Solution:

Six sided die marked ' 2 ' on one face, ' 3 ' on two faces and ' 4 ' on three faces.

When it is thrown twice, we get 36 sample points.

' X ' denotes sum of the face numbers and the possible values of ' X ' are 4,5,6,7 and 8

For $X=4$, the sample point is (2,2)

For $X=5$, the sample points are (2,3),(3,2)

For $X=6$, the sample points are (3,3),(2,4),(4,2)

For $X=7$, the sample points are (3,4),(4,3)

For $X=8$, the sample point is (4,4)

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Ex 11.2**Question 1.**

Three fair coins are tossed simultaneously. Find the probability mass function for number of

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heads occurred.

Solution:

When three coins are tossed, the sample space is

$$S=\{HHH,HHT,HHT,HTH,HTT,THH,THT,TTT\}$$

' X ' is the random variable denotes the number of heads.

Therefore ' X ' can take the values of 0,1,2 and 3

Hence, the probabilities

$$P(X=0)=P(\text{No}$$

$$\text{heads})=1/8; P(X=1)=P(1 \text{ head})=3/8; P(X=2)=P(2 \text{ heads})=3/8; P(X=3)=P(3 \text{ heads})=1/8$$

Question 2.

A six sided die is marked ' 1 ' on one face, ' 3 ' on two of its faces, and ' 5 ' on remaining three faces. The die is thrown twice. If X denotes the total score in two throws, find

(i) the probability mass function

(ii) the cumulative distribution function

(iii) $P(4$

$$X < 10)$$

(iv) $P(X$

6)

Solution:

Given that die is marked ' 1 ' on one face, ' 3 ' on two of its faces and ' 5 ' on remaining three faces. i.e., {1, 3,3,5,5,5} in a single die.

When it is thrown twice, the number of sample points is 36, in which the sum of faces numbers are 2,4,6, 8 and 10 are the value of random variable ' X '.

$$P(X=2)=1/36; P(X=4)=4/36; P(X=6)=10/36; P(X=8)=12/36; P(X=10)=9/36$$

(i) Probability mass function:

(ii) The Cumulative distribution function:

$$F(x)=\sum_{x_i \leq x} P(X=x_i)$$

$$F(2)=1/36$$

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$\frac{1}{36}$ for $x < 21/36$ for x

$25/36$

for x

$415/36$

for x

$627/36$

for x

81

for x

10

(iii) (4

$10) = P(X=4) + P(X=6) + P(X=8)$

$= \frac{436}{36} + \frac{1036}{36} + \frac{1236}{36} = \frac{2636}{36} = 1318$

(iv) $P(X$

$6) = P(X=6) + P(X=8) + P(X=10) = \frac{1036}{36} + \frac{1236}{36} + \frac{936}{36} = \frac{3136}{36}$

Question 3.

Find the probability mass function and cumulative distribution function of number of girl child in families with 4 children, assuming equal probabilities for boys and girls.

Solution:

Let 'X' be the random variable which denotes the number of girl children in the family of 4 children and X takes the values of 0,1,2,3,4.

Probability of child being a boy $= P(B) = \frac{1}{2}$

Probability of child being a girl $= P(G) = \frac{1}{2}$

$P(X=0) = P(4 \text{ Boys}) = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$ $P(X=1) = P(1G, 3B) = \binom{4}{1} \left(\frac{1}{2}\right)^4 = \frac{4}{16} = \frac{1}{4}$

$P(X=2) = P(2G, 2B) = \binom{4}{2} \left(\frac{1}{2}\right)^4 = \frac{6}{16} = \frac{3}{8}$ $P(X=3) = P(3G, 1B) = \binom{4}{3} \left(\frac{1}{2}\right)^4 = \frac{4}{16} = \frac{1}{4}$

$P(X=4) = P(4 \text{ Girls}) = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$ The probability mass function

is $f(x) = \frac{1}{16}$

$\frac{1}{16}$

$\frac{1}{16}$ for $x=0, \frac{1}{4}$ for $x=1, \frac{3}{8}$ for $x=2$ Cumulative distribution function $F(x) = \frac{1}{16}$

$\frac{1}{16}$

$\frac{1}{16}$ for $x < 0$ $\frac{1}{16}$ for x

$05/16$

for x

$111/16$

for x

$215/16$

for x

31

for x

4

Question 4.

Suppose a discrete random variable can only take the values 0,1, and 2. The probability mass function is defined by $f(x) = \frac{x+1}{k}$, for $x=0,1,2$ otherwise

Find

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(i) the value of k

(ii) cumulative distribution function

(iii) $P(X$

1)

Solution:

Given probability mass function $f(x) = \frac{x+1}{k}$, for $x=0,1,2$, otherwise

(i) Value of k $\sum_{x=0}^2 f(x) = f(0) + f(1) + f(2) = 1$

i.e., $1k+2k+5k=1$

(ii) Cumulative distribution function: $F(x)=$

0 for $x < 0$ 1/8 for x

0.3/8

for x

11

for x

2

(iii) $P(X$

$1)=P(X=1)+P(X=2)=28+58=78$

Question 5.

The cumulative distribution function of a discrete random variable is given by

$F(x)=$

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0

$x <$

10.15

1

$x < 0.35$ $x < 10.60$ $x < 20.85$ $x < 31.3$ $x <$

Find the (i) the probability mass function

(ii) $P(X < 1)$

(iii) $P(X$

2)

Solution:

Given cumulative distribution function is $F(x)=$

0

$x <$

10.15

1

$x < 0.35$ $x < 10.60$ $x < 20.85$ $x < 31.3$ $x <$

The values of 'x' are

1, 0, 1, 2, 3

nis $\mathrm{F}(x)=\left\{\begin{array}{cr} 0 & -\infty \text{ For } \mathrm{x}=-1, \mathrm{f}(\mathrm{x})=0.15-0=0.15 \text{ For } \mathrm{x}=0, \mathrm{f}(\mathrm{x})=0.35-0.15=0.20 \text{ For } \mathrm{x}=1, \mathrm{f}(\mathrm{x})=0.60-0.35=0.25 \text{ For } \mathrm{x}=2, \mathrm{f}(\mathrm{x})=0.85-0.60=0.25 \text{ For } \mathrm{x}=3, \mathrm{f}(\mathrm{x})=1-0.85=0.15 \end{array}\right.$

(i) Probability mass function table

(ii) $P(X < 1)=P(X=$

$1)+P(X=0)=0.15+0.20=0.35$

(iii) $P(X$

2)=

$P(X=2)+P(X=3)=0.25+0.15=0.40$

Question 6.

A random variable X has the following probability mass function.

Solution:

Given probability mass function

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(i) We know that $\sum P_i=1$

i.e., $k^2+2k^2+3k^2+2k+3k=1$

$6k^2+5k=1$

$$6k^2 + 5k$$

$$1 = 0$$

$$(k+1)(6k$$

$$1) = 0$$

$$(ii) P(2$$

$$X < 5) = P(X=2) + P(X=3) + P(X=4) = 2k$$

$$2 + 3k^2 + 2k = 5k^2 + 2k = 5(136) + 2(16) = 536 + 13$$

$$= 1736$$

$$(iii) P(3 < X) = P(X > 3)$$

$$= P(X=4) + P(X=5) = 2k + 3k = 5k = 5/6$$

Question 7.

The cumulative distribution function of a discrete random variable is given by

Find (i) the probability mass function

$$(ii) P(X < 3) \text{ and}$$

$$(iii) P(X$$

$$2)$$

Solution:

Given cumulative distribution function is

(i) The value of 'x' are 0, 1, 2, 3 and 4

$$\text{For } x=0, f(x) = 12$$

$$0 =$$

$$12; \text{ For } x=1, f(x) = 35 - 12 = 110$$

$$\text{For } x=2, f(x) = 45 - 35 = 15; \text{ For } x=3, f(x) = 910 - 45 = 110$$

$$\text{For } x=4, f(x) = 1 - 910 = 110$$

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Probability mass function is

$$(ii) P(X < 3) = P(X=0) + P(X=1) + P(X=2) = 12 + 110 + 15 = 5 + 1 + 210 = 810 = 45$$

$$(iii) P(X$$

$$2) =$$

$$P(X=2) + P(X=3) + P(X=4) = 15 + 110 + 110 = 2 + 210 = 410 = 25$$

Ex 11.3

Question 1.

The probability density function of X is given by

$$f(x) = \begin{cases} kxe^{-x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$$

$$2$$

$$-x \text{ for } x > 0 \text{ for } x$$

Find the value of k

Solution:

$$f(x) = \begin{cases} kxe^{-x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$$

$$2$$

$$-x \text{ for } x > 0 \text{ for } x$$

Since, f(x) is a Probability density function

$$\text{i.e., } \int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^{\infty} kxe^{-x} dx = 1$$

$$2$$

$$-x dx = 1/k$$

$$\int_{-\infty}^{\infty} kxe^{-x} dx = 1$$

$$2$$

$$-x dx = 1$$

Using, integration by parts method

Question 2.

The probability density function of X is $f(x) = \begin{cases} kx & \text{for } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$ Solution:

$P(0.2)$

$$\int_{0.6}^{0.2} x dx = [x^2/2]_{0.6}^{0.2} = 12[0.36 - 0.04] = 12[0.32] = 3.84$$

(ii)

$$X < 1.8) =$$

$$\int_{1.81}^{2} (2-x) dx = [2x - \frac{x^2}{2}]_{1.81}^2 = (3.6$$

1.62)

(2.4

$$0.72) = 1.98$$

$$1.68 = 0.$$

30

(iii)

$P(0.5)$

$$X < 1.5) =$$

$$\int 10.5x dx + \int 1.51(2-x) dx = \left[x^2 \right] 10.5 + \left[2x - x^2 \right] 1.51 = 0.375 + 0.375 = 0.75$$

Suppose the amount of milk sold daily at a milk booth is distributed with a minimum of 200 litres and a maximum of 600 litres with probability density function

$$f(x) = \begin{cases} k200 - x \end{cases}$$

6000

otherwise

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Find (i) the value of k

(ii) the distribution function

(iii) the probability that daily sales will fall between 300 litres and 500 litres?

Solution:

Given p.d.f is $f(x) = \begin{cases} k, & 200 \leq x \end{cases}$

6000,

otherwise

(i) Since $f(x)$ is a p.d.f, $\int_{-\infty}^{\infty} f(x) dx = 1$

i.e.,

$$k \int_{0.6}^{0.8} 600200 dx = 1k[x]_{0.6}^{0.8} 600200 = 1k(600$$

$$200)=1400$$

$$k=1 \rightarrow k=1400$$

(ii) Now, the probability density function $f(x) = \{1/400,200 \times x$

6000.

otherwise Integrating

$f(x)$, we get probability density function

$$F(x) = \begin{cases} 0 & \text{for } x < 200 \\ x/400 & \text{for } 200 \leq x \end{cases}$$

6000

for $x > 600$

(iii) The probability that daily sales will fall between 300 litres and 500 litres is

$$\int_{500}^{300} f(x) dx = 1400 \int_{500}^{300} dx = 1400 [x]_{500}^{300} = 1400 [500$$

$$300] = 200400 = 12$$

The probability density function of X is given by $f(x) = \begin{cases} ke^{-x^3} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$

Find (i) the value of k

(ii) the distribution function

(iii) $P(X < 3)$

(iv) $P(5)$

(v) $P(X$

4)

Solution:Given probability density function $f(x) = \begin{cases} ke^{-x/3} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$ (i) Since $f(x)$ is a p.d.f, $\int_{-\infty}^{\infty} f(x) dx = 1$

i.e.,

$$\int_{-\infty}^{\infty} ke^{-x/3} dx = 1$$

$$\int_{-\infty}^{\infty} e^{-x/3} dx = 1/k$$

$$[-3e^{-x/3}]_{-\infty}^{\infty} = 1/k$$

$$-3[0 - e^0] = 1/k$$

$$-3[0 - 1] = 1/k$$

$$3 = 1/k$$

$$k = 1/3$$

$$f(x) = \begin{cases} \frac{1}{3}e^{-x/3} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$$

$$f(x) = \frac{1}{3}e^{-x/3}$$

$$f(x) = \frac{1}{3}e^{-x/3}$$

(ii) Probability density function $f(x) = \begin{cases} \frac{1}{3}e^{-x/3} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$ Integrating $f(x)$, we getProbability density function $F(x) = \begin{cases} \frac{1}{3}e^{-x/3} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$

$$F(x) = \frac{1}{3}e^{-x/3}$$

for $x > 0$ Therefore $F(x) = \begin{cases} \frac{1}{3}e^{-x/3} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$

(iii)

$$P(X < 3) = \int_{-\infty}^3 f(x) dx = \int_{-\infty}^3 \frac{1}{3}e^{-x/3} dx = -[e^{-x/3}]_{-\infty}^3 = -[e^{-3/3} - e^0] = 1 - e^{-1}$$

$$= 1 - e^{-1}$$

$$= 1 - e^{-1}$$

$$= 1 - e^{-1}$$

(iv)

$$P(X < 5)$$

$$= \int_{-\infty}^5 f(x) dx$$

$$= \int_{-\infty}^5 \frac{1}{3}e^{-x/3} dx = -[e^{-x/3}]_{-\infty}^5 = -[e^{-5/3} - e^0] = 1 - e^{-5/3}$$

$$= 1 - e^{-5/3}$$

$$= 1 - e^{-5/3}$$

$$= 1 - e^{-5/3}$$

$$= 1 - e^{-5/3}$$

(v)

$$P(X < 4)$$

$$= \int_{-\infty}^4 f(x) dx$$

$$= \int_{-\infty}^4 \frac{1}{3}e^{-x/3} dx = -[e^{-x/3}]_{-\infty}^4 = -[e^{-4/3} - e^0] = 1 - e^{-4/3}$$

$$= 1 - e^{-4/3}$$

$$= 1 - e^{-4/3}$$

$$= 1 - e^{-4/3}$$

$$= 1 - e^{-4/3}$$

Question 5.If X is a random variable with probability density function $f(x)$ given by,

$$f(x) = \begin{cases} kx+1 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f(x) = kx+1$$

$$f(x) = \begin{cases} kx+1 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Solution:

Given probability density function is

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otherwise

(i) Integrating $f(x)$ and separating with respect to the intervals we get, Probability distribution

function

$f(x)$ is a Probability density function.

$$0$$

$$1$$

$$-(x+1)dx = \frac{1}{2}(x^2+x) \Big|_0^1$$

$$1$$

$$-1 + c = \frac{1}{2} + c = 1 \Rightarrow c = \frac{3}{2}$$

$$\frac{1}{2} = \frac{1}{2}$$

(ii) $P($

$$0.5$$

$$0.5) =$$

$$0$$

$$0.5$$

$$(x+1)dx + \frac{1}{2}(-x+1)dx = \frac{1}{2}(x^2+x) \Big|_0^1$$

$$0.5$$

$$[-\frac{x^2}{2} + x] \Big|_{0.5}^1 =$$

$$(0.125$$

$$0.5)] +$$

$$0.125 + 0.5] =$$

$$0.25 = 0.75$$

Question 6.

If X is the random variable with probability density function $F(x)$ given by,

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{2}(x^2 + x), & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

$$1$$

then find (i) the distribution function $f(x)$

(ii) $P(0.3$

$$0.6)$$

Solution:

Given probability distribution function $F(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{2}(x^2 + x), & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$

$$1$$

(i) Differentiating $F(x)$, we get $f(x)$

$$\text{i.e., Probability density function } f(x) = F'(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{2}(2x+1), & 0 \leq x < 1 \\ 0, & x \geq 1 \end{cases}$$

$$1$$

(ii)

$$P(0.3$$

$$0.6) = \frac{1}{2}$$

$$\int_{0.3}^{0.6} \frac{1}{2}(2x+1)dx = \frac{1}{2} \left[x^2 + x \right]_{0.3}^{0.6} = \frac{1}{2} [(0.36 + 0.6)$$

$$(0.09 + 0.3)] = \frac{1}{2}(0.$$

$$57) = 0.285$$

Ex 11.4

Question 1.

For the random variable X with the given probability mass function as below, find the mean and variance

$$(i) f(x) = \begin{cases} \frac{1}{10}, & x = 2, 5 \\ \frac{1}{5}, & x = 0, 1, 3, 4 \end{cases}$$

$$(ii) f(x) = \begin{cases} 4-x, & x = 1, 2, 3 \end{cases}$$

$$(iii) f(x) = \begin{cases} 2(x-1) & \& 1 \end{cases} \text{ (iv) } f(x) = \begin{cases} 12e^{-x/2} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Solution:

$$(i) \text{ Given probability mass function } f(x) = \begin{cases} \frac{1}{10}, & x = 2, 5 \\ \frac{1}{5}, & x = 0, 1, 3, 4 \end{cases}$$

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$$\text{Mean } E(X) = \sum x f(x)$$

$$= 0 \cdot \frac{1}{5} + 1 \cdot \frac{1}{5} + 3 \cdot \frac{1}{5} + 4 \cdot \frac{1}{5} + 2 \cdot \frac{1}{10} + 5 \cdot \frac{1}{10} = \frac{9}{5} + \frac{12}{10} = \frac{18}{10} + \frac{6}{10} = \frac{24}{10} = 2.4$$

$$E(X^2) = \sum x^2 f(x)$$

$$= 0 \cdot \frac{1}{5} + 1 \cdot \frac{1}{5} + 9 \cdot \frac{1}{5} + 16 \cdot \frac{1}{5} + 4 \cdot \frac{1}{10} + 25 \cdot \frac{1}{10}$$

$$=15+25+95+165+52=285+52=8110$$

$$\text{Variance } \text{Var}(X)=E(X^2)$$

$$[E(X)]$$

$$2=8110$$

$$529100=810$$

$$529100=281100=2.81$$

(ii) Given probability mass function $f(x)=\{4-x^6, x=1,2,3$

$$\text{Mean } E(X)=\sum x f(x)=12+23+12=1+23=53=1.67 E(X^2)=\sum x^2 f(x)=12+43+96=3+8+96=20$$

$$6=103$$

$$\text{Variance } \text{Var}(X)=E(X^2)$$

$$[E(X)]$$

$$2=103-(53)^2=103$$

$$259=30$$

$$259=59=0.56$$

(iii) Given probability mass function $f(x)=\begin{cases} 2(x-1), & 1 \leq x \leq 2 \\ 1-x, & 2 < x \leq 3 \end{cases}$ Here ' X ' is a continuous random variable

$$\text{Mean } E(X)=\int_1^2 x f(x) dx + \int_2^3 x f(x) dx = 2 \int_1^2 x(x-1) dx + \int_2^3 x(1-x) dx$$

$$= 2 \left[\frac{x^3}{3} - \frac{x^2}{2} \right]_1^2 + \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_2^3$$

$$= 2 \left[\frac{8}{3} - \frac{1}{3} - \frac{4}{2} + \frac{1}{2} \right] + \left[\frac{9}{2} - \frac{8}{3} - \frac{4}{2} + \frac{8}{3} \right]$$

$$= 2 \left[\frac{7}{6} - \frac{3}{6} \right] + \left[\frac{9}{6} - \frac{8}{6} - \frac{4}{6} + \frac{8}{6} \right] = 2 \left[\frac{4}{6} \right] + \left[\frac{1}{6} \right] = \frac{8}{6} + \frac{1}{6} = \frac{9}{6} = 1.5$$

$$2$$

$$-)-(-$$

$$13$$

$$12$$

$$=2[23+16]=2(56)=53 E(X^2)=\int_1^2 x^2 f(x) dx + \int_2^3 x^2 f(x) dx = 2 \int_1^2 x^2(x-1) dx + \int_2^3 x^2(1-x) dx$$

$$1)$$

$$dx=2 \int_1^2 (x^3-x^2) dx = 2 \left[\frac{x^4}{4} - \frac{x^3}{3} \right]_1^2 + \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_2^3$$

$$= 2 \left[\frac{16}{4} - \frac{1}{4} - \frac{8}{3} + \frac{1}{3} \right] + \left[\frac{27}{3} - \frac{16}{4} - \frac{8}{3} + \frac{16}{4} \right] = 2 \left[\frac{15}{4} - \frac{7}{3} \right] + \left[\frac{27}{3} - \frac{16}{4} - \frac{8}{3} + \frac{16}{4} \right]$$

$$[E(X)]$$

$$2=176-(53)^2=176$$

$$259=51$$

$$5018=118$$

(iv)

$$f(x)=\begin{cases} 1/2e^{-x/2}, & \text{for } x>0 \\ 0, & \text{otherwise} \end{cases}$$

Since ' X ' is a continuous random variable

Mean

$$E(X)=\int_0^{\infty} x f(x) dx = \int_0^{\infty} x \cdot \frac{1}{2} e^{-x/2} dx$$

$$= \frac{1}{2} \int_0^{\infty} x e^{-x/2} dx = \frac{1}{2} \left[-2x e^{-x/2} - 2 e^{-x/2} \right]_0^{\infty}$$

$$= \frac{1}{2} \left[0 - (-2 \cdot 0 - 2) \right] = \frac{1}{2} \cdot 2 = 1$$

$$-xe^{-x/2}$$

$$4$$

$$-e^{-x/2}]$$

$$= -[xe^{-x/2} + 2e^{-x/2}]_0^{\infty}$$

$$2]$$

$$[0$$

$$(0+\infty)=2$$

(Using integration by parts)

$$\text{Let } u=x, dv=e^{-x/2}$$

$$dv=e^{-x/2} dx \Rightarrow v=-2e^{-x/2}$$

$$1/2$$

$$u dv = uv - \int v du = x(-2e^{-x/2}) - \int (-2e^{-x/2}) dx$$

$$= -2xe^{-x/2} + 2 \int e^{-x/2} dx$$

$$= -2xe^{-x/2} + 2(-2e^{-x/2}) + C$$

$$= -2xe^{-x/2} - 4e^{-x/2} + C$$

$$= -2xe^{-x/2} - 4e^{-x/2} + C$$

$$2$$

$$1/2 =$$

$$2$$

$$-xe^{-x/2}$$

$$4$$

$$-e^{-x/2}$$

[Since Using integration parts twice]

$$= -[x^2 e^{-x/2} + 4x e^{-x/2} + 8e^{-x/2}]$$

$$(0$$

$$8) = 8$$

$$\text{Variance } \text{Var}(X) = E(X^2)$$

$$[E(X)]$$

$$2 = 8$$

$$4 = 4$$

Question 2.

Two balls are drawn in succession without replacement from an urn containing four red balls and three black balls. Let X be the possible outcomes drawing red balls. Find the probability mass function and mean for X .

Solution:

Number of Red balls = 4

Number of Black balls = 3

Total number of balls = 7

Given : two balls are drawn in succession without replacement.

Let ' X ' be the number of red balls and ' X ' can take the values 0, 1 and 2.

$$P(X=0) = P(BB) = \frac{3C_2}{7C_2} = \frac{1}{7} \quad P(X=1) = P(RB)$$

$$+ P(BR) = \left(\frac{4C_1 3C_1}{7C_2} \right) + \left(\frac{3C_1 4C_1}{7C_2} \right) = \frac{4}{7} \quad P(X=2) = P(RR) = \frac{4C_2}{7C_2} = \frac{2}{7}$$

-- Source page 15 --

Therefore Probability mass function

$$E(X) = \sum x f(x) = 0 + \frac{4}{7} + \frac{4}{7} = \frac{8}{7}$$

Question 3.

If μ and σ^2 are the mean and variance of the discrete random variable X , and $E(X+3) = 10$ and $E(X+3)^2 = 116$, find μ and σ^2 .

Solution:

Mean = μ

Variance = σ^2

$$\text{Variance } \text{Var}(X) = E(X^2)$$

$$[E(X)]$$

$$265$$

$$49 = 16 =$$

$$\sigma^2 \text{ Therefore } \mu = 7 \text{ and } \sigma^2 = 16$$

Question 4.

Four fair coins are tossed once. Find the probability mass function, mean and variance for number of heads occurred.

Solution:

Let ' X ' be the number of heads occurred when four coins are tossed once. Hence ' X ' can take the values 0, 1, 2, 3 and 4.

When 4 coins are tossed, the sample space is,

$$S = \{ HHHH, HHHT, HHTH, HHTT, HTHH, HTHT, HTTH, HTTT, THHH, THHT, THTH, THTT, TTHH, TTHT, TTTH, TTTT \}$$

$$P(X=0) = \frac{1}{16}, P(X=1) = \frac{4}{16}, P(X=2) = \frac{6}{16}, P(X=3) = \frac{4}{16}, P(X=4) = \frac{1}{16}.$$

Probability mass function

$$\text{Mean } E(X) = \sum x f(x) = 0 + 1 + 2 + 3 + 4 = 10$$

$$E(X^2) = \sum x^2 f(x) = 0 + 1 + 4 + 9 + 16 = 30 \quad \text{Variance } \text{Var}(X) = E(X^2) -$$

$$[E(X)]^2 = 5$$

$$4 = 1$$

Question 5.

A commuter train arrives punctually at a station every half hour. Each morning, a student leaves his house to the train station. Let X denote the amount of time, in minutes, that the student waits for the train from the time he reaches the train station. It is known that the p.d.f of X is.

$$f(x) = \begin{cases} 1/30, & 0 \leq x \leq 30 \\ 0, & \text{elsewhere} \end{cases}$$

Obtain and interpret the expected value of the random variable X .

Solution:

Given, p.d.f is $f(x) = \begin{cases} 1/30, & 0 \leq x \leq 30 \\ 0, & \text{elsewhere} \end{cases}$ is a continuous random variable.

-- Source page 16 --

Therefore Expected value

$$\text{of } X = E(X) = \int_0^{30} x f(x) dx = \int_0^{30} x \cdot \frac{1}{30} dx = \frac{1}{30} \left[\frac{x^2}{2} \right]_0^{30} = \frac{1}{30} \cdot \frac{30^2}{2} = \frac{1}{30} \cdot \frac{900}{2} = \frac{1}{30} \cdot 450 = 15 \text{ minutes}$$

Question 6.

The time to failure in thousands of hours of an electronic equipment used in a manufactured

$$f(x) = \begin{cases} 3e^{-3x}, & x \geq 0 \\ 0, & \text{elsewhere} \end{cases}$$

$$f(x) = \begin{cases} 3e^{-3x}, & x \geq 0 \\ 0, & \text{elsewhere} \end{cases}$$

Find the expected life of this computer has the density function. electronic equipment.

Solution:

Given p.d.f is $f(x) = \begin{cases} 3e^{-3x}, & x \geq 0 \\ 0, & \text{elsewhere} \end{cases}$

$$f(x) = \begin{cases} 3e^{-3x}, & x \geq 0 \\ 0, & \text{elsewhere} \end{cases}$$

Expected life

$E(X) = \int_0^{\infty} x f(x) dx$

$$= \int_0^{\infty} x \cdot 3e^{-3x} dx$$

$$= 3 \int_0^{\infty} x e^{-3x} dx$$

$$= 3 \int_0^{\infty} x e^{-3x} dx$$

$$= 3 \int_0^{\infty} x e^{-3x} dx$$

$$= 3 \int_0^{\infty} x e^{-3x} dx$$

(Using integration by parts method)

$$\text{Let } u = x \text{ and } dv = e^{-3x} dx$$

$$du = dx \text{ and } v = -\frac{1}{3} e^{-3x}$$

$$= 3 \int_0^{\infty} x e^{-3x} dx$$

$$= 3 \left[-\frac{x}{3} e^{-3x} - \int -\frac{1}{3} e^{-3x} dx \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$= 3 \left[-\frac{x}{3} e^{-3x} + \frac{1}{9} e^{-3x} \right]_0^{\infty}$$

$$1/3)$$

$$[xe$$

$$3$$

$$-x+e$$

$$3$$

$$-x^3]$$

$$\square = -[0 - (0 + 13)] = 13$$

Only $E(X)$ is required.

Question 7.

The probability density function of the random variable X is given by

$$f(x) = \begin{cases} 16xe^{-x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$$

$$4$$

$$-x \text{ for } x > 0 \text{ for } x$$

$\square$ Find the mean and variance of X .

Solution:

Given p.d.f. is $f(x) = \begin{cases} 16xe^{-x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$

$$4$$

$$-x \text{ for } x > 0 \text{ for } x$$

$$\text{Mean } E(X) = \int_{-\infty}^{\infty} xf(x)dx = 16 \int_0^{\infty} x^2 e^{-x} dx$$

$$\int_0^{\infty} x^2 e^{-x} dx$$

$$4$$

$$-x dx$$

(Using integration by parts method twice)

$$\text{Let } u = x^2 \quad dv = e^{-x} dx$$

$$\text{and } du = 2x dx \quad v = -e^{-x}$$

$$4$$

$$-x, v = e^{-x}$$

$$4$$

$$4$$

[Since Integration by parts method]

$$u = x^2 \quad dv = dx$$

$$\text{and } du = 2x dx \quad v = e^{-x}$$

$$4$$

$$-x dx, v = e^{-x}$$

$$4$$

$$4$$

$$\int_0^{\infty} u dv = uv - \int_0^{\infty} v du = -xe^{-x}$$

$$4$$

$$-x dx = -xe^{-x}$$

$$4$$

$$-x^4 + 14e^{-x}$$

$$4$$

$$-x dx = -xe^{-x}$$

$$4$$

$$-x^4$$

$$116$$

$$4$$

Substituting in (1)

$$\int_0^{\infty} x^2 e^{-x} dx$$

$$4$$

$$-x dx = x^2 e^{-x}$$

$$4$$

$$4 + 12$$

$$[-xe^{-x}]_0^{\infty}$$

4

 $-x^4$

116

4

 $-x]E(X)=16[x^2e$

4

4

 $--xe$

4

 $-x^8-e$

4

 $-x^32]$ $0=16[0$ $1/32]=16(1/32)=1/2E$ $(X^2)=\int_0^4 x^2 f(x) dx = 16 \int_0^4$ $x^3 e$

4

 $-x dx$ *(Using integration by parts method)*Let, $u = x^3 \Rightarrow du = 3x^2 dx$ and $dv = e$

4

 $-x dx \Rightarrow v = e$

4

4

 $\int u dv = uv - \int v du \Rightarrow x^3 e$

4

 $-x dx = -x^3 e$

4

 $-x^4 + 34 \int e$

4

 $-x \int x^2 dx = -x^3 e$

4

 $-x^4 + 34 [x^2 e$

4

4

 $--xe$

4

 $x^8 - e$

4

 $-x^32]$ *[Since Using E(X) integration]* $= -x^3 e$

4

 $-x^4 - 316 x^2 e$

4

 $-x - 332 x e$

4

 $-x - 3128 e$

4

 $E(X^2) = 16[-x^3 e$

4

 $-x^4$

316

 $x^2 e$

4

332

4

3128

4

-x]

 $\bar{x}=16[0-$ $3/128)]=16$ $(3128)=38$ Variance $\text{Var}(X)=E(X^2)$ $[E(X)]$

2

-- Source page 17 --

 $=38-14=3$

28

--18

Question 8.

A lottery with 600 tickets gives one prize of 200, four prizes of 100, and six prizes of 50. If the ticket costs is 2, find the expected winning amount of a ticket.

Solution:

Given, total number of tickets =600

Prizes to be given : One prize of Rs. 200

Four prizes of Rs. 100

Six prizes of Rs. 50

Let 'X' be the random variable denotes the winning amount and it can take the values 200,100 and 50.

Probability of winning Rs. 200=1/600

Probability of winning Rs. 100=4/600

Probability of winning Rs. 50=6/600

Therefore Probability mass function is

Therefore $E(X)=200 \times \frac{1}{600} + 100 \times \frac{4}{600} + 50 \times \frac{6}{600} = 1.5$

Expected winning amount = Amount won - Cost of lottery
=1.50

2.00=

0.50

i.e., Loss of Rs. 0.50

Ex11.5**Question 1.**Compute $P(X=k)$ for the binomial distribution, $B(n,p)$ where(i) $n=6, p=1/3, k=3$ (ii) $n=10, p=1/5, k=4$ (iii) $n=9, p=1/2, k=7$ **Solution:**(i) Given $n=6, p=1/3, k=3$ Therefore $q=1-p=2/3$ $13=23P(X=$ $x)=nC_x p^x q^{n-x}, x=0,1,2,\dots,n$ $nP(X=k)=P(X=3)=6C_3 (1/3)^3 (2/3)^3$

3

 $=20(1/27)(8/27)=160/729$ (ii) $n=10, p=1/5, k=4$ Therefore $q=1-p=4/5$

$$15 = 45P(X=4) = nCk p^k q^{n-k} = {}^{45}C_4 (15/45)^4 (30/45)^{41}$$

$$= {}^{45}C_4 (1/3)^4 (2/3)^{41}$$

$$= \frac{45!}{4!41!} \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^{41}$$

$$= \frac{45 \times 44 \times 43 \times 42 \times 41!}{24 \times 41!} \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^{41}$$

$$= \frac{45 \times 44 \times 43 \times 42}{24} \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^{41}$$

$$= 15 \times 11 \times 43 \times 7 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^{41}$$

$$= 15 \times 11 \times 43 \times 7 \times \frac{1}{81} \times \frac{2^{41}}{3^{41}}$$

$$= \frac{15 \times 11 \times 43 \times 7 \times 2^{41}}{3^{42}}$$

Question 2.

The probability that Mr. Q hits a target at any trial is $\frac{1}{4}$. Suppose he tries at the target 10 times.

Find the probability that he hits the target

- (i) exactly 4 times
- (ii) at least one time.

Solution:

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Let 'p' be the probability of hitting the trial

i.e., $p = \frac{1}{4}$, Therefore $q = 1 - p = 1 - \frac{1}{4} = \frac{3}{4}$

and number of trials 'n' = 10

Probability of 'x' success in 'n' trials is

$$P(X=x) = nCk p^k q^{n-k}, x=0,1,2,\dots,n$$

(i) Probability that Mr. Q hits the target exactly 4 times is

$$P(X=4) = {}^{10}C_4 \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^6$$

$$= \frac{10!}{4!6!} \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^6$$

(ii) Probability that Mr. Q hits the target at least one time is

$$P(X \geq 1)$$

$$= 1 - P(X=0)$$

$$= 1 - P(X=0)$$

$$= 1 - {}^{10}C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^{10}$$

$$= 1 - \left(\frac{3}{4}\right)^{10}$$

$$= 1 - \left(\frac{3}{4}\right)^{10}$$

$$= 1 - \left(\frac{3}{4}\right)^{10}$$

$$= 1 - \left(\frac{3}{4}\right)^{10}$$

$$= 1 - \left(\frac{3}{4}\right)^{10}$$

$$= 1 - \left(\frac{3}{4}\right)^{10}$$

Question 3.

Using binomial distribution find the mean and variance of X for the following experiments

- (i) A fair coin is tossed 100 times, and X denote the number of heads.
- (ii) A fair die is tossed 240 times, and X denote the number of times that four appeared.

Solution:

(i) $n=100$, 'X' denotes the number of heads.

Therefore Probability of getting a head $p=\frac{1}{2}$ and $q=1-p=\frac{1}{2}$

$$\text{Mean} = np = 100 \times \frac{1}{2} = 50 \quad \text{Variance} = npq = 100 \times \frac{1}{2} \times \frac{1}{2} = 25$$

(ii) $n=240$, 'X' denotes the number of times four appeared.

Therefore Probability of getting four $p=\frac{1}{6}$ and $q=1-p=\frac{5}{6}$

$$\text{Mean} = np = 240 \times \frac{1}{6} = 40; \quad \text{Variance} = npq = 240 \times \frac{1}{6} \times \frac{5}{6} = 100\frac{2}{3}$$

Question 4.

The probability that a certain kind of component will survive a electrical test is $\frac{3}{4}$. Find the probability that exactly 3 of the 5 components tested survive.

Solution:

Given $n=5$

Probability that a component survive in a test $=p=3/4$

Therefore $q=1-p=1-3/4=1/4$

Let 'X' be the random variable denotes the number of components survived in a test.

Probability of 'x' successes in 'n' trials is.

$$P(X=x) = {}^nC_x p^x q^{n-x}, x=0,1,2,\dots,n$$

Probability that exactly 3 components survive

$$P(X=3) = {}^5C_3 (3/4)^3 (1/4)^2$$

3

$$= {}^5C_3 (3/4)^3 (1/4)^2 = 270/1024$$

Question 5.

A retailer purchases a certain kind of electronic device from a manufacturer. The manufacturer indicates that the defective rate of the device is 5%. The inspector of the retailer randomly picks 10 items from a shipment. What is the probability that there will be

(i) at least one defective item

(ii) exactly two defective items.

Solution:

Given $n=10$

Probability of a defective item $=p=5\%=0.05$

Therefore $q=1-p=1-0.05=0.95$

Let 'X' be the random variable denotes the number of defective items.

-- Source page-19 --

Therefore Probability of 'x' successes in 'n' trials is

$$P(X=x) = {}^nC_x p^x q^{n-x}, x=0,1,2,\dots,n$$

(i) Probability that atleast one defective item will be there

$$P(X$$

$$1) = 1$$

$$P(X < 1) = 1$$

$$[P(X=0)] = 1 - [{}^{10}C_0 (0.05)^0 (0.95)^{10}]$$

$$= 1 - (0.95)^{10}$$

$$(0.95)$$

$$10$$

(ii) Probability that exactly two defective item will be there

$$P(X=2) = {}^{10}C_2 (0.05)^2 (0.95)^8 = {}^{10}C_2 (0.05)^2 (0.95)^8$$

Question 6.

If the probability that a fluorescent light has a useful life of at least 600 hours is 0.9, find the probabilities that among 12 such lights.

(i) exactly 10 will have a useful life of at least 600 hours;

(ii) at least 11 will have a useful life of at least 600 hours;

(iii) at least 2 will not have a useful life of at least 600 hours.

Solution:

Given $n=12$

Probability that a fluorescent light has a life of atleast of 600 hours is $p=0.9$

Therefore $q=1$

$$p=1$$

$$0.9=0.1$$

Let 'X' be the number of bulbs.

Therefore Probability of 'x' successes in 'n' trials is

$$P(X=x) = {}^nC_x p^x q^{n-x}, x=0,1,2,\dots,n$$

(i) Probability that exactly 10 bulbs will have a useful life of atleast 600 hours

$$P(X=10) = {}^{12}C_{10} (0.9)^{10} (0.1)^2$$

(ii) Probability that atleast 11 will have a useful life of atleast 600 hours is

$$P(X$$

$$11) = P(X=11) + P(X=12) = 12C$$

$$11(0.9)11(0.1)1 + 12C12(0.9)12(0.1)0 = 12(0.9)11(0.1) + (0.$$

$$9)12(1) = (0.9)11(12 + 0.9) = (2.1)(0.9)11$$

(iii) Probability that atleast 2 will not have a useful life of 600 hours is
 $= 1 - P(X$

$$2) =$$

$$P(X < 2) = P(X=0) + P(X=1) = 12C0(0.9)0(0.1)12 + 12C1(0.9)1(0.1)11 = (0.1)12 + 12($$

$$0.9)(0.1)11 = (0.1)11(0.1 + 10.8) = (10.9)(0.1)11$$

Question 7.

The mean and standard deviation of a binomial variate X are respectively 6 and 2. Find

(i) the probability mass function

(ii) $P(X=3)$

(iii) $P(X$

2)

Solution:

Given, Mean $= np = 6$ Standard deviation $= \sqrt{npq} = 2$

Substituting (1) in (2)

$$6$$

$$\sqrt{q} = 4 \Rightarrow q = 16 = 23 \text{ Therefore } p = 1 - q = 1 - 23 = 13 \quad np = 6 \Rightarrow n(13) = 6 \Rightarrow n = 18$$

Probability of 'x' success in 'n' trials is

$$P(X=x) = nC_x p^x q^{n-x}, x=0, 1, 2, \dots, n$$

(i) Probability mass function

$$P(X=x) = 18C_x (13)^x (23)^{18-x}$$

$$(ii) P(X=3) = 18C_3 (13)^3 (23)^{15}$$

(iii)

$$P(X$$

$$2) = 1$$

$$- P(X < 2) = 1$$

$$- P(X=0) + P(X=1) = 1 - [18C_0 (13)^0 (23)^{18} + 18C_1 (13)^1 (23)^{17}] = 1 -$$

$$[(23)^{18} + 18(13)(23)^{17}] = 1 - (23)^{17} [23 + 6] = 1$$

$$203$$

$$(23)^{17}$$

Question 8.

If X

$B(n, p)$

such that $4P(X=4) = P(X=2)$ and $n=6$. Find the distribution, mean and standard deviation.

-- Source page 20 --

Solution:

n, p i.e., 'X' is the binomial variate given $n=6$.

$$P(X=x) = nC_x p^x q^{n-x}, x=0, 1, 2, \dots$$

$$n4P(X=4) = P(X=2) \Rightarrow 6C_4 p^4 q^2 = 6C_2 p^2 q^4 \Rightarrow p^2 = q^2 \Rightarrow (1-p)^2 = p^2 \Rightarrow 1 - 2p + p^2 = p^2$$

$$1 = 0$$

$$p^2 = 2p$$

$$1$$

$$= 0(3p$$

$$1)($$

$$p+1=0 \text{ Therefore } p=1/3; p=$$

$$1$$

- is not possible. If $p=1/3$ then $q=2/3$

$$1/3 = 2/3$$

Binomial Distribution is $B(6, 1/3)$

$$\text{Mean } np = 6 \cdot 1/3 = 2 \text{ Standard deviation } = \sqrt{npq} = \sqrt{6 \cdot 1/3 \cdot 2/3} = \sqrt{4/3} = 2/\sqrt{3}$$

Question 9.

In a binomial distribution consisting of 5 independent trials, the probability of 1 and 2 successes are 0.4096 and 0.2048 respectively. Find the mean and variance of the distribution.

solution:

Number of trials $n=5$

Probability of 'x' successes in 'n' trials is

$$P(X=x) = nC_x p^x q^{n-x}, x=0,1,2,\dots,n$$

Given $P(X=1)=0.4096$ and $P(X=2)=0.2048$

$$P(X=1) = 5C_1 p^1 q^{5-1}$$

$$5C_1$$

$$1p^1q^4 = 5p^1q^4 = 0.4096 \quad P(X=2) = 0.2048$$

$$5C_2$$

$$2p^2q^3 = 10p^2q^3 = 0.2048$$

Dividing Eq.(1) by Eq.(2)

$$\frac{5p^1q^4}{10p^2q^3} = \frac{0.4096}{0.2048} \quad \frac{q}{2p} = 2 \quad q = 4p$$

$$1$$

$$q - p = 4p \quad [\text{Since } q = 1 - p]$$

$$5$$

$$q = 1$$

Ex 11.6

Choose the Correct or the most suitable answer from the given four alternatives:

Question 1.

Let X be random variable with probability density function

$$f(x) = \begin{cases} 2/x^3 & 10 \\ x < 1 \end{cases}$$

$$10$$

$$x < 1$$

Which of the following statement is correct

(a) both mean and variance exist

(b) mean exists but variance does not exist

(c) both mean and variance do not exist

(d) variance exists but Mean does not exist

Solution:

(b) mean exists but variance does not exist

Hint:

Given probability density function $f(x) = \begin{cases} 2/x^3 & 10 \\ x < 1 \end{cases}$

$$10$$

$$x < 1$$

$$\text{Mean} = E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^1 x \cdot \frac{2}{x^3} dx = \int_0^1 \frac{2}{x^2} dx = \left[-\frac{2}{x} \right]_0^1 = -2 - (-\infty) = \infty$$

$$1$$

$$\int_0^1 \frac{2}{x^2} dx = \left[-\frac{2}{x} \right]_0^1 = -2 - (-\infty) = \infty$$

$$x^3 dx = \frac{x^4}{4} \Big|_0^1 = \frac{1}{4} - 0 = \frac{1}{4}$$

$$1$$

$$\int_0^1 \frac{2}{x^2} dx = \left[-\frac{2}{x} \right]_0^1 = -2 - (-\infty) = \infty$$

$$(2/$$

$$1$$

$$(0$$

$$2) = 2E$$

$$(X^2) = \int_0^1 x^2 f(x) dx = \int_0^1 x^2 \cdot \frac{2}{x^3} dx = \int_0^1 \frac{2}{x} dx = 2 \ln x \Big|_0^1 = 2(0 - (-\infty)) = \infty$$

$$1$$

$$x^2 f(x) dx = \int_0^1 \frac{2}{x} dx = 2 \ln x \Big|_0^1 = 2(0 - (-\infty)) = \infty$$

Question 2 .

A rod of length 21 is broken into two pieces at random. The probability density function of the

shorter of the two pieces is $f(x) = \begin{cases} \frac{1}{l} & 0 \leq x \leq l \\ 0 & \text{otherwise} \end{cases}$. The mean and variance of the shorter of the two pieces are respectively

- (a) $l/2, l^2/3$
- (b) $l/2, l^2/6$
- (c) $l/2, l^2/12$
- (d) $l/2, l^2/24$

Solution:

- (d) $l/2, l^2/24$

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Hint:

$$f(x) = \begin{cases} \frac{1}{l} & 0 \leq x \leq l \\ 0 & \text{otherwise} \end{cases} \quad \text{Mean} = E(X) = \int_0^l x \cdot \frac{1}{l} dx = \frac{1}{l} \left[\frac{x^2}{2} \right]_0^l = \frac{1}{l} \cdot \frac{l^2}{2} = \frac{l}{2}$$

$$E(X^2) = \int_0^l x^2 \cdot \frac{1}{l} dx = \frac{1}{l} \left[\frac{x^3}{3} \right]_0^l = \frac{1}{l} \cdot \frac{l^3}{3} = \frac{l^2}{3}$$

$$\text{Variance} = \text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{l^2}{3} - \left(\frac{l}{2}\right)^2 = \frac{l^2}{3} - \frac{l^2}{4} = \frac{l^2}{12}$$

Question 3.

Consider a game where the player tosses a six-sided fair die. If the face that comes up is 6, the player wins Rs. 36, otherwise he loses Rs. k , where k is the face that comes up $k \in \{1, 2, 3, 4, 5\}$.

The expected amount to win at this game is

- (a) 196
- (b) -196
- (c) 32
- (d) -32

Solution:

- (b) -196

Hint:

$$\text{Expected amount to win} = 1 \cdot 36 + (-k) \cdot \frac{5}{6} = -196$$

Question 4.

A pair of dice numbered 1, 2, 3, 4, 5, 6 of a six-sided die and 1, 2, 3, 4 of a four-sided die is rolled and the sum is determined. Let the random variable X denote this sum. Then the number of elements in the inverse image of 7 is

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Solution:

- (d) 4

Hint:

Sample points to get the sum ' 7 ' are (3,4), (4,3), (5,2), (6,1)
Number of elements in inverse image of 7 = 4

Question 5.

A random variable X has binomial distribution with $n=25$ and $p=0.8$ then standard deviation of X is

- (a) 6
- (b) 4
- (c) 3
- (d) 2

Solution:

- (d) 2

Hint:

$$n=25, p=0.8, \text{Therefore } q=1$$

$$0.8=0.2$$

$$\text{Standard deviation} = \sqrt{npq} = \sqrt{25 \times 0.8 \times 0.2} = \sqrt{4} = 2$$

Question 6.

Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed n times. Then the possible values of X are

(a) $i+2n, i=0,1,2,\dots,n$

(b) $2i$

$n, i=0,1,2,\dots,n$

(c) n

$i, i=0,1,2,\dots,n$

(d) $2i+2n, i=0,1,2,\dots,n$

Solution:

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(b) $2i$

$n, i=0,1,2,\dots,n$

Question 7.

If the function $f(x)=112$ for a (a) 0 and 12

(b) 5 and 17

(c) 7 and 19

(d) 16 and 24

Solution:

(d) 16 and 24

Hint:

Since $f(x)$ is a probability density function, $\int_a^b f(x)dx=1$

But when $a=16$ and $b=24$, $\int_{16}^{24} 16f(x)dx=1$

Question 8.

Four buses carrying 160 students from the same school arrive at a football stadium. The buses carry, respectively, 42, 36, 34, and 48 students. One of the students is randomly selected. Let X denote the number of students that were on the bus carrying the randomly selected student. One of the 4 bus drivers is also randomly selected. Let Y denote the number of students on that bus. Then $E[X]$ and $E[Y]$ respectively are

(a) 50, 40

(b) 40, 50

(c) 40.75, 40

(d) 41, 41

Solution:

(c) 40.75, 40

Hint:

$E[X] > E[Y]$. Since the students selected is more likely to come from the bus carrying maximum number of students whereas the drivers are equally likely to be selected.

$$E[X] = (42 \times 42/160) + (36 \times 36/160) + (34 \times 34/160) + (48 \times 48/160) = 40.75$$

$$E[Y] = (14 \times 42) + (14 \times 36) + (14 \times 34) + (14 \times 48) = 40$$

Question 9.

Two coins are to be flipped. The first coin will land on heads with probability 0.6, the second with Probability 0.5. Assume that the results of the flips are independent, and let X equal the total number of heads that result. The value of $E[X]$ is

(a) 0.11

(b) 1.1

(c) 11

(d) 1

Solution:

(b) 11

Hint:

$$P(X=0)=P(TT)=P(T)P(T)=(0.4)(0.5)=0.20$$

$$P(X=1)=P(HT)+P(TH)=(0.6)(0.5)+(0.4)(0.5)=0.5$$

$$P(X=2)=P(HH)=P(H).P(H)=(0.6)(0.5)=0.30$$

$$E[X]=\sum xP(x)=(0)(0.20)+(1)(0.5)+(2)(0.30)=0.5+0.6=1.1$$

Question 10.

On a multiple-choice exam with 3 possible destructives for each of the 5 questions, the probability that a student will get 4 or more correct answers just by guessing is

(a) 11243

(b) 38

(c) 1243

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(d) 5243

Solution:

(a) 11243

Hint:

$$p=13, q=23 \text{ and } n=5$$

$$P(X=x)=nC_x p^x q^{n-x}, x=0,1,2,\dots,5$$

$$nP(X=4)+P(X=5)=5C_4(13)^4(23)^1+5C_5(13)^5(23)^0$$

$$=5(13)^4(23)+(13)^5=(13)^4[5(23)+13]$$

$$=1135=11243$$

Question 11.If $P\{X=0\}=1$ $P\{X=1\}$. If $E[X]=3\text{Var}(X)$, then $P\{X=0\}$.

(a) 23

(b) 25

(c) 15

(d) 13

Solution:

(d) 13

Hint:Given $P(X=0)=1$ $P(X=1)$ $P(X=1)=1$ $P(X=0)$ Let $n=P(X=0)$, Therefore $P(X=1)=1$ $E(X)=(0)(n)+(1)(1$ $n)=1$ $E(X^2)=(0)^2(n)+(1)^2(1$ $n)=1$ Now substituting $E(X)=3\text{Var}(X)$, we get $n=13$ (or) 1Therefore $P(X=0)=13$ **Question 12.**If X is a binomial random variable with expected value 6 and variance 2.4, Then $P\{X=5\}$ is(a) $(105)(35)6(25)4$ (b) $(105)(35)5$ (c) $(105)(35)4(25)6$ (d) $(105)(35)5(25)5$ **Solution:**

(d) $(105)(35)5(25)5$

Hint:

Given, $np=6$ and $npq=2.4$

$q=npqnp=2.46=0.4$ Therefore $p=1-q=1$

$0.4=0.6$

$np=6n(0.6)=6 \times 6=10$ $P(X=5)=10C5(61$

$0)5(410)5=10C5(35)5(25)5$

Question 13.

The random variable X has the probability density function $f(x)=\left\{\begin{array}{cc}$

$a x+b, \& 0 \leq x \leq 1$ and $\frac{1}{2}(b)\frac{1}{2}\$ and 1$

(c) 2 and 1

(d) 1 and 2

Solution:

(a) 1 and 12

Hint:

Given probability density function $f(x)=\left\{\begin{array}{cc}$

$a x+b, \& 0 \leq x \leq 1$ Since, $f(x)$ is a probability density function, $\int_0^1 (a x+b) dx=1$

$\int_0^1 (a x+b) dx=1$ $a x+b=1$ $a x+b=2$ $E(X)=\int_0^1 x f(x) dx=\int_0^1 x (a x+b) dx=\int_0^1 (a x^2+b x) dx=\frac{a x^3}{3}+\frac{b x^2}{2}\bigg|_0^1=\frac{a}{3}+\frac{b}{2}$

$=\frac{a}{3}+\frac{b}{2}$

$=\frac{2}{3}$

$\frac{a}{3}+\frac{b}{2}=\frac{2}{3}$

Solving these equations, we get $a=1$ and $b=\frac{1}{2}$

Question 14.

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Suppose that X takes on one of the values 0, 1, and 2. If for some constant k ,

$P(X=i)=kP(X=i$

1)

for $i=1,2$, $P(X=0)=1/7$. Then the value of k is

(a) 1

(b) 2

(c) 3

(d) 4

Solution:

(b) 2

Hint:

$P(X=i)=kP(X=i$

1)

for $i=1,2$ and $P(X=0)=1/7$

Solving the equation, we get $k=2$

Question 15.

Which of the following is a discrete random variable?

I. The number of cars crossing a particular signal in a day.

II. The number of customers in a queue to buy train tickets at a moment.

III. The time taken to complete a telephone call.

(a) I and II

(b) II only

(c) III only

(d) II and III

Solution:

(a) I and II

Question 16.

$f(x) = \begin{cases} 2x, & 0 \leq x \leq a \\ 0, & \text{otherwise} \end{cases}$ is a probability density function of a random variable, then the value of a is

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Solution:

- (a) 1

Hint:

Since, $f(x)$ is a probability density function, $\int_0^a f(x) dx = 1$

$$\int_0^a 2x dx = 1 \Rightarrow [x^2]_0^a = 1 \Rightarrow a^2 = 1 \Rightarrow a = 1$$

Question 17.

The probability function of a random variable is defined as

Then $E(X)$ is equal to

- (a) 115
- (b) 110
- (c) 13
- (d) 23

Solution:

- (d) 23

Hint:

$$E(X) = \sum x f(x) = \sum x P_i = 1 \cdot E(X) =$$

2

2

$$-k + 0 + 4k + 10k \text{ Therefore } 15k = 1 \Rightarrow k = \frac{1}{15} = 0.0667 \Rightarrow 23k = 115$$

Question 18.

Let X have a Bernoulli distribution with mean 0.4, then the variance of $(2X$

3)

is

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- (a) 0.24
- (b) 0.48
- (c) 0.6
- (d) 0.96

Solution:

- (d) 0.96

Hint:

$$\text{Mean} = 0.4 \Rightarrow p = 0.4 \Rightarrow \text{Var}(2X$$

3) = 2

$$2 \text{Var} X = 4(pq) = 4(0.4)(0.6) = 0.96$$

Question 19.

If in 6 trials, X is a binomial variate which follows the relation $9P(X=4) = P(X=2)$, then the probability of success is ...

- (a) 0.125
- (b) 0.25
- (c) 0.375
- (d) 0.75

Solution:

- (b) 0.25

Hint:

$$n=6, P(X=x) = {}^nC_x p^x q^{n-x}, x=0,1,2,\dots,n$$

$$\text{Given, } {}^9P(X=4) = P(X=2)$$

$${}^9C_4 p^4 q^2 = {}^9C_2 p^2 q^7 \quad \text{[Since } {}^nC_4 = {}^nC_2 \text{]} \quad 9p^2 = (1$$

$$29p^2 = 1$$

$$2p = 1$$

$$28p^2 + 2p$$

$$1 = 0(4p$$

$$1)$$

$$(2p+1) = 0 \quad p = 14,$$

$$12$$

$$[p =$$

$$12$$

$$\text{is not possible }] p = 14 = 0.25$$

Question 20.

A computer salesperson knows from his past experience that he sells computers to one in every twenty customers who enter the showroom. What is the probability that he will sell a computer to exactly two of the next three customers?

$$(a) \frac{57}{203}$$

$$(b) \frac{57}{202}$$

$$(c) \frac{193}{203}$$

$$(d) \frac{57}{20}$$

Solution:

$$(a) \left[\frac{57}{203} \right]$$

$$\{ \frac{57}{203} \} \quad [/ \text{ latex }]$$

Hint:

$$p = \frac{1}{20}, q = \frac{19}{20} \text{ and } n = 3 \quad P(X=x) = {}^nC_x p^x q^{n-x}, x=0,1,2,\dots,n$$

$$n P(X=2) = {}^3C_2 \left(\frac{1}{20} \right)^2 \left(\frac{19}{20} \right)^1 = \frac{3(19)(1)}{(20)^3} = \frac{57}{20^3}$$